



Politecnico di Milano

# Dipartimento di Elettronica, Informazione e Bioingegneria

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## Parallel Computing–I part–Thursday, June 13<sup>th</sup>, 2024

**Polimi ID** \_\_\_\_\_

**Surname** \_\_\_\_\_ **Name** \_\_\_\_\_

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

**Exercise 1 (4 points)** \_\_\_\_\_

**Exercise 2 (4 points)** \_\_\_\_\_

**Exercise 3 (4 points)** \_\_\_\_\_

**Exercise 3 (4 points)** \_\_\_\_\_

### Exercise n. 1

Answer the following questions about PRAM models and briefly explain (without an explanation, the answer will be considered invalid)

- A. Please list the characteristics of the PRAM model that simplify the space complexity estimations (1)

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- B. Could you please list the properties/features PRAM has in common with CUDA? (2)

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- C. What does it mean "fixed time model"? Is it related to Amdahl or Gustafson? (1)

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## Exercise n. 2

Answer the following questions about different forms of parallel executions and briefly explain (without an explanation, the answer will be considered invalid).

A. Prefetching reduces stalls. True/False? Explain what is meant by prefetching and stall. (1)

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B. Please explain what is meant by "Bandwidth bound application."(1)

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C. Explain how the impact of the stalls could be reduced. (1)

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D. Please compare Fine-grain multithreading with Simultaneous multithreading. (1)

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### Exercise n. 3

Answer the following questions about programming models plus CUDA and briefly explain (without an explanation, the answer will be considered invalid).

A. What is meant by the term Warp in CUDA? Is it part of the CUDA language? (1)

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B. CUDA kernels may create dependencies between threads in a block. True/False? (1)

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C. Describe the distinct types of address spaces visible to kernels. How many? How much is shared? How fast are they? (2)

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#### Exercise n. 4

Answer the following questions about memory and heterogeneous systems and briefly explain (without an explanation, the answer will be considered invalid).

- A. Please briefly describe what Memory coalescing means. (1) **This question can be skipped in case of a passed challenge.**

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- B. Please briefly describe what DRAM burst means. (1)

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- C. DRAM Banking could reduce the dead time. True/False? Why? (1)

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- D. Please list the memory operation orderings in a sequentially consistent memory system. (1)

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